**Project Proposal — Hotel Booking Cancellation Prediction Using Machine Learning**

**Date:** July 2026

**1. Certificate Name**

**Maryan Ahmed Warsame**

---

**2. Project Title and Description**

**Title:**

**Hotel Booking Cancellation Prediction Using Machine Learning**

Hotels receive thousands of booking requests, and many reservations are canceled, which affects revenue and resource planning. This project aims to develop a machine learning system that predicts whether a hotel booking will be canceled based on customer details, booking information, and reservation characteristics.

The project will apply supervised learning algorithms to predict booking cancellations and analyze customer booking patterns using machine learning techniques.

**3. Problem Type**

This project uses:

| Learning Type | Problem | Goal |
|---|---|---|
| Supervised Learning | Binary Classification | Predict if a booking will be canceled or not |
| Unsupervised Learning | Clustering | Discover groups of customers with similar booking behaviors |

**Target Column:** is_canceled

| Value | Meaning |
|---|---|
| 0 | Not canceled |
| 1 | Canceled |

**4. Dataset**

**Dataset Name:** Hotel Booking Demand Dataset

**Source:** Kaggle
https://www.kaggle.com/datasets/pattae/hotel-booking-demand-datasets

**Information Details**

| | |
|---|---|
| Rows | 119,390 |
| Columns | 32 |
| Data Type | Structured Tabular Data |
| Target | is_canceled |

**Main Features:**

| CATEGORY | EXAMPLES |
|---|---|
| **CUSTOMER DETAILS** | adults, children, country, customer type |
| **BOOKING DETAILS** | lead_time, stay duration, arrival information |
| **HOTEL DETAILS** | hotel, meal, reserved_room_type |
| **PREVIOUS HISTORY** | previous_cancellations, previous_bookings_not_canceled |
| **FINANCIAL DATA** | adr, deposit_type |

---

**5. Data Preprocessing Plan**

The following preprocessing steps will be applied:

| **Step** | **Description** |
|---|---|
| Data Inspection | Understand dataset structure and data types |
| Missing Values | Handle missing values in relevant columns |
| Duplicate Removal | Remove duplicated records |
| Outlier Handling | Analyze and manage extreme values |
| Encoding | Apply One-Hot Encoding for categorical variables |
| Scaling | Scale numerical features |

| Step | Description |
|---|---|
| Data Splitting | Split data into training and testing sets (80/20) |

**Encoding Method: One-Hot Encoding**

One-Hot Encoding is selected because most categorical variables do not have a natural order.

**6. Algorithms I Plan to Train**

The following classification algorithms will be trained and compared:

| Algorithm | Reason |
|---|---|
| Logistic Regression | Baseline model for binary classification and easy interpretation |
| Random Forest Classifier | Handles complex relationships and reduces overfitting |
| Gradient Boosting Classifier | Strong ensemble algorithm for tabular datasets |

For customer behavior analysis, clustering algorithms will also be applied:

| Algorithm | Purpose |
|---|---|
| K-Means Clustering | Groups customers based on similarities |
| Agglomerative Clustering | Creates hierarchical customer groups |

**7. Evaluation Plan**

**Classification Metrics:**

| Metric | Purpose |
|---|---|
| Accuracy | Overall prediction performance |
| Precision | Correct cancellation predictions |
| Recall | Detecting actual cancellations |
| F1-Score | Balance between precision and recall |
| Confusion Matrix | Shows prediction errors |

| Metric | Purpose |
|---|---|
| ROC-AUC | Measures model performance |

The best model will be selected based on F1-score.

**Clustering Evaluation:**

- Elbow Method
- Silhouette Score

**8. Deployment Plan**

The best trained model can be deployed using **FastAPI**.

The API will accept booking information and return a cancellation prediction with probability.

Example:

Input:

{

"hotel": "City Hotel",

"lead_time": 120,

"adults": 2,

"adr": 150

}

Output:

{

"prediction": "Canceled",

"probability": 0.82

}

**9. Repository Plan**

```
hotel-booking-ml-project/
├── dataset/
|   └── hotel_bookings.csv
```

```
├── notebooks/
|   ├── 01_data_cleaning_eda.ipynb
|   └── 02_model_training.ipynb
├── src/
|   ├── preprocess.py
|   └── train.py
├── models/
|   └── best_model.pkl
├── reports/
|   └── project_report.pdf
├── README.md
├── project_proposal.md
└── requirements.txt
```

**Conclusion**

This project applies supervised and unsupervised machine learning techniques to real-world hotel booking data. The classification models will predict booking cancellations, while clustering methods will provide additional insights into customer booking patterns.

The project demonstrates skills in data preprocessing, exploratory data analysis, machine learning model development, evaluation, and deployment.